Tallinn University of Technology

Manipulator Workspace (WS)

Author: Jordan Thomas Brodie
Student Code: 243483
Individual variant: RPR8
Course: ATR0030 Robotics

October 19, 2026

1 Task

The purpose of this work was to plot the collision-free workspace of my assigned manipulator, **variant RPR8**, in the xyz Cartesian coordinates, together with its views in the xy, xz and yz planes.

The workspace was found with the brute-force (generate and test) method. The forward kinematics model from the FK exercise was solved repeatedly while the revolute joints changed their angles A and B cyclically and the prismatic joint changed its length L_2 within set limits. After each iteration the end-effector (EE) position was tested for collisions with the floor, and only collision-free positions were saved in a grab array. The saved array was then used to plot the workspace and its three plane views.

2 Initial Data

My assigned manipulator is the **RPR8** variant (Revolute – Prismatic – Revolute). The link dimensions are the same as in the FK model and are shown in Table 1.

Table 1: Initial link dimensions for variant RPR8

Parameter	Value	Description
L_0	3	Base offset
L_1	5	J1–J2 link length
$L_{2,\max}$	6	Maximum prismatic displacement
L_3	5	J3–EE link length

The variable parameters of the manipulator are:

- A – angle of the first revolute joint J1, rotating around the local z-axis of J1;
- L_2 – length of the prismatic link between J2 and J3, along the local $-z$-axis of J2;
- B – angle of the final revolute joint J3, rotating around the local z-axis of J3.

At the home state $A = B = 0$ and the prismatic link has its maximum length $L_2 = L_{2,\max} = 6$, which gives the FK result $P_E = \begin{bmatrix} 6 & 5 & 8 \end{bmatrix}$.

3 Model Parameters

The fixed geometry is taken directly from the verified RPR8 forward kinematics model. The constant translation vectors are

$$T_{01} = \begin{bmatrix} 0 & 0 & L_0 \end{bmatrix},$$
$$T_{12} = \begin{bmatrix} L_1 & 0 & 0 \end{bmatrix},$$

and

$$T_{3E} = \begin{bmatrix} 0 & L_3 & 0 \end{bmatrix}.$$

The J0 frame is rotated around its y-axis by $-\pi/2$ to obtain the J1 frame:

$$R_{01} = \begin{bmatrix} 0 & 1 & 0 & -\pi/2 \end{bmatrix}.$$

In the workspace model the three joint variables are inserted into this chain as an axis-angle rotation for each revolute joint and a translation for the prismatic joint:

$$r_1 = \begin{bmatrix} 0 & 0 & 1 & A \end{bmatrix}, \qquad t_2 = \begin{bmatrix} 0 & 0 & -L_2 \end{bmatrix}, \qquad r_3 = \begin{bmatrix} 0 & 0 & 1 & B \end{bmatrix}.$$

All vectors are converted to homogeneous matrices with `trvec2tform()` and `axang2tform()`, and the transformations are combined in sequence:

$$^0T_1 = T_{01}\, R_{01}\, r_1,$$

$$^0T_2 = {}^0T_1\, T_{12},$$

$$^0T_3 = {}^0T_2\, t_2,$$

and

$$^0T_E = {}^0T_3\, r_3\, T_{3E}.$$

The EE position P_E is the translation part of ^0T_E, extracted with `tform2trvec()`.

4 Joint Limits and Collision Avoidance

Because J1 is rotated by $-\pi/2$ around y, its local z-axis is horizontal (along the base $-x$-axis). Angle A therefore swings link L_1 up and down in the yz plane around the point J1 at height $L_0 = 3$. Without limits, J2, J3, link L_3 and the EE would hit the floor.

The lowest point reached by the EE during one full revolution of B is

$$z_{E,\min} = L_0 + L_1 \cos A - L_3.$$

To keep the EE above the floor with the tolerance $d = 0.05$ for *every* angle B, the condition $z_{E,\min} \geq d$ must hold, which gives

$$\cos A \geq \frac{L_3 - L_0 + d}{L_1} = \frac{5 - 3 + 0.05}{5} = 0.41 \quad \Rightarrow \quad |A| \leq 1.148 \text{ rad} \approx 65.8^\circ.$$

The resulting joint limits used in the program are listed in Table 2.

Table 2: Joint limits used in the workspace search

Joint variable	Minimum	Maximum	Reason
A	-1.148 rad	1.148 rad	EE and J3 stay above the floor
B	0	2π	full revolution is collision-free
L_2	0	6	prismatic stroke

Each range was divided into $N = 20$ steps, so the program solved the forward kinematics $20^3 = 8000$ times. In every iteration two collision checks were performed: the EE height $P_E(3) < d$ and the J3 height $P_3(3) < d$. Only collision-free EE positions were added to the grab array `poses`. With the limits above, all 8000 configurations passed both checks (the lowest EE height was $z \approx 0.065$ and the lowest J3 height $z \approx 5.05$), so the program reported *"No collisions detected."*

5 MATLAB Program

The complete MATLAB program used to generate the RPR8 workspace is shown below.

```
clear; clc; close all

%% From JordanBrodie_FK.m
% Link lengths - RPR8
L0=3; L1=5; L2max=6; L3=5;
```

```
% Offset and reorientation vectors - RPR8
% Same geometry as the verified RPR8 Forward Kinematics model

% J0 -> J1: translation along base +z
T01=[0 0 L0];

% Rotate J0 frame -pi/2 around y-axis to obtain J1 frame
R01=[0 1 0 -pi/2];

% J1 -> J2: translation along local +x
T12=[L1 0 0];

% J3 -> EE: translation along local +y
T3E=[0 L3 0];

% Their homogeneous matrices
T01H=trvec2tform(T01);
R01H=axang2tform(R01);
T12H=trvec2tform(T12);
T3EH=trvec2tform(T3E);

%% Tolerance and joint limits
d=0.05; % tolerance

% Joint limits to prevent floor collision
% A is limited so J3, L3 and EE remain above floor
% for the complete 360 degree rotation of B
minA=-1.148;
maxA= 1.148;

% B can complete one full revolution
minB=0;
maxB=2*pi;

% RPR8 prismatic joint L2
minL=0; maxL=L2max;

% Number of cycles and ranges
N=20;

rangeA=linspace(minA, maxA, N);
rangeB=linspace(minB, maxB, N);
rangeL=linspace(maxL, minL, N);

% Container of EE poses and message
poses=[]; msg='';

%% Jordan Brodie RPR8 kinematics model
for L2=rangeL % or 0 to stop L
for B=rangeB  % or 0 to stop B
for A=rangeA  % or 0 to stop A

% Jordan Brodie variable A,B,L parameters
% RPR8 = Revolute - Prismatic - Revolute

% First revolute joint A
% Rotation around the local z-axis of J1
r1=[0 0 1 A];
```

```
% Prismatic joint L2
% In the home FK model J2 -> J3 is along local -z
t2=[0 0 -L2];

% Final revolute joint B
% Rotation around the local z-axis of J3
r3=[0 0 1 B];

% Convert variable parameters to homogeneous matrices
r1h=axang2tform(r1);
t2h=trvec2tform(t2);
r3h=axang2tform(r3);

% RPR8 model considering variable parameters
%
% Home model from FK:
% J1 = T01H * R01H
% J2 = J1 * T12H
% J3 = J2 * T23H
% EE = J3 * T3EH
%
% Workspace model inserts A, L2 and B into this chain.

% First revolute joint
J1=T01H * R01H * r1h;

% Fixed link L1
J2=J1 * T12H;

% Prismatic joint L2
J3=J2 * t2h;

% Final revolute joint B and fixed link L3
EE=J3 * r3h * T3EH;

% Joint and EE positions
P1=tform2trvec(J1);
P2=tform2trvec(J2);
P3=tform2trvec(J3);
PE=tform2trvec(EE);


%% Collecting EE positions

% Checking if EE collides with floor
collision=PE(3)<d;

if collision

    msg=sprintf( ...
        'EE can collide with floor at z=%.2f, A=%.2f, B=%.2f, L2=%.2f', ...
        PE(3),A,B,L2);

else

    % Checking if J3 goes below the floor
    collision=P3(3)<d;
```

```
    if collision

        msg=sprintf( ...
            'J3 can collide with floor at z=%.2f, A=%.2f, B=%.2f, L2=%.2f', ...
            P3(3),A,B,L2);

    end
end

% Save only collision-free EE positions
if ~collision
    poses=[poses, PE(:)];
end

end
end
end

%% Functions that can be called to explore workspace
if isempty(msg)
    disp('No collisions detected.')
else
    disp('Collision configurations were detected and removed from the workspace.')
    disp(msg)
end

x=poses(1,:);
y=poses(2,:);
z=poses(3,:);

% 3D workspace
figure;
windowFunction;
title('Jordan Brodie RPR8 workspace')
surface([x; x], [y; y], [z; z], ...
    'EdgeColor', 'interp')
view(45,30)


% XY view
figure;
windowFunction;
title('Jordan Brodie RPR8 XY view')
surface([x; x], [y; y], [z; z], ...
    'EdgeColor', 'interp')
view(2)

% XZ view
figure;
windowFunction;
title('Jordan Brodie RPR8 XZ view')
surface([x; x], [y; y], [z; z], ...
    'EdgeColor', 'interp')
view(1,1)

% YZ view
figure;
```

```
windowFunction;
title('Jordan Brodie RPR8 YZ view')
surface([x; x], [y; y], [z; z], ...
    'EdgeColor', 'interp')
view(90,1)

%% ============================================================
% USER-DEFINED FUNCTIONS
% ============================================================

%% Figure window
function windowFunction

plot3(0,0,0,'o')

hold on
grid on
axis equal

xlabel('x')
ylabel('y')
zlabel('z')

end
```

6 Simulation Results

Figure 1 shows the collision-free 3D workspace of the RPR8 manipulator generated by the MATLAB program. Figures 2a–3 show its views in the xy, xz and yz planes.

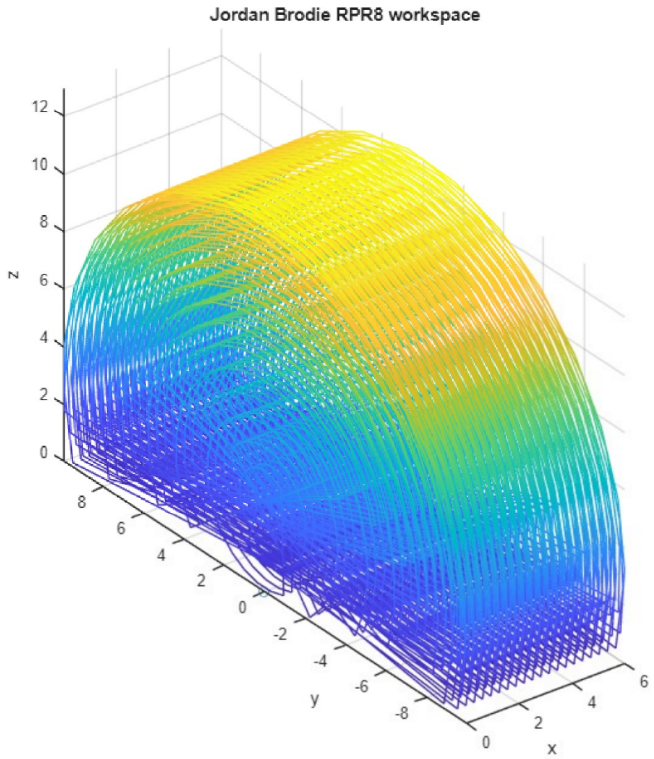
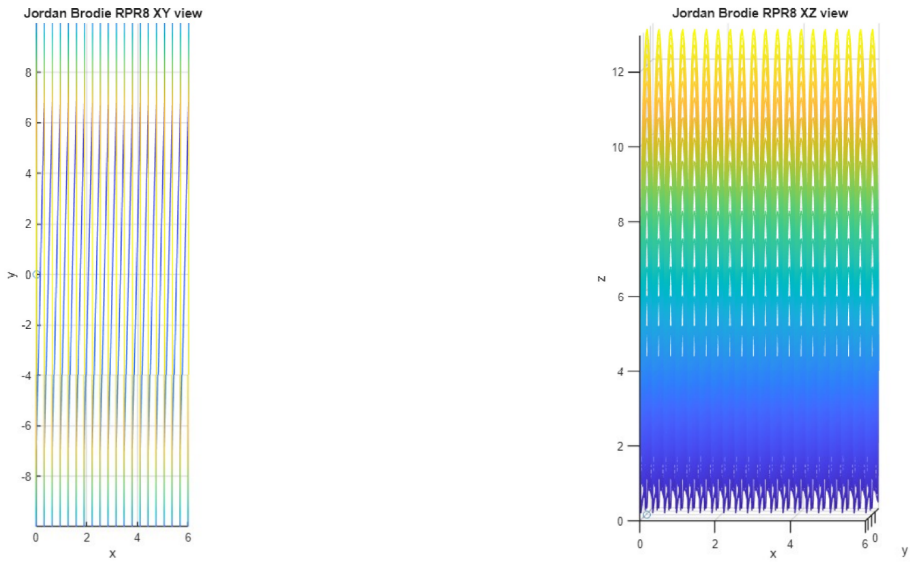


Figure 1: MATLAB plot of the RPR8 collision-free workspace.



(a) XY view. (b) XZ view.

Figure 2: Top (XY) and side (XZ) views of the RPR8 workspace.

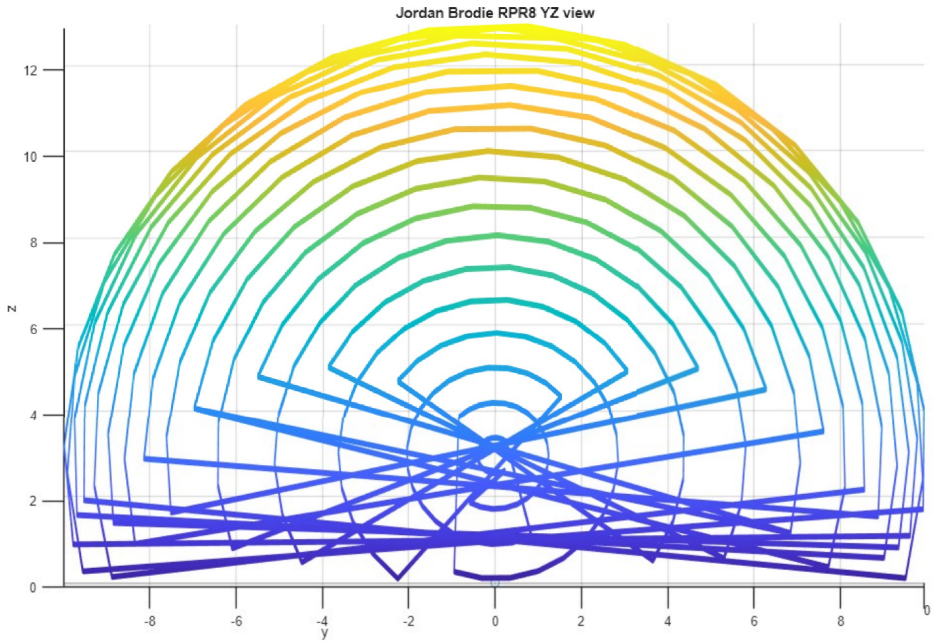


Figure 3: Front (YZ) view of the RPR8 workspace.

The plots agree with the geometry of the RPR8 manipulator:

- The **XY** and **XZ** views show that the workspace extends from $x = 0$ to $x = 6$. This x-range is produced only by the prismatic joint L_2, because L_2 is the only link that moves along the base x-axis.
- The **YZ** view shows the shape created by the two revolute joints. Rotation A moves J2 and J3 on an arc of radius $L_1 = 5$ around J1, and rotation B moves the EE on a full circle of radius $L_3 = 5$ around J3. Together they give a dome-shaped region from $y \approx -9.6$ to $y \approx 9.5$ and from just above the floor up to $z_{\max} = L_0 + L_1 + L_3 = 13$, which is reached when $A = 0$ and the EE points straight up.

- The 3D view combines both effects: the yz dome is extruded along x between 0 and 6, forming a half-cylinder-like workspace that never goes below the floor.

The straight lines that cross the inside of the YZ view are not part of the workspace boundary. They appear because `surface()` connects consecutive points of the grab array, so the last point of one B cycle is joined to the first point of the next one.

7 Conclusions

In this assignment I used my RPR8 forward kinematics model to find and plot the collision-free workspace of the manipulator in MATLAB.

The hardest part for me was choosing the joint limits. At first it was not obvious which joint could cause a collision, because the J1 frame is rotated and angle A rotates around a horizontal axis instead of a vertical one. Once I understood that A lifts and lowers the whole arm in the yz plane, I could write the lowest EE height as a function of A and calculate the limit $|A| \leq 1.148$ rad, which keeps the EE above the floor even when B makes a full revolution.

This exercise also showed me how each joint contributes to the shape of the workspace: the prismatic joint stretches it along x, and the two revolute joints define its dome-shaped cross-section in the yz plane. The brute-force method is simple to program, but its accuracy depends on the number of cycles N, and a finer grid would make the boundary smoother at the cost of longer computation.

The final program found 8000 collision-free EE positions and produced the expected workspace, which confirms that the transformations from the FK model and the chosen joint limits were set up correctly.